

DUSTINE TRIEU

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SUMMARY

Computer Science senior at the University of Delaware with experience building backend systems, data pipelines, and full-stack applications. Developed REST APIs, GPU-accelerated processing pipelines, and modular backtesting infrastructure. Seeking entry-level software engineering roles focused on backend and data-intensive systems.

EDUCATION

University of Delaware - Newark, DE

May 2026

B.S. Computer Science, B.A. Asian Studies; Minor in Mathematics

SKILLS

Languages: Python, C++, TypeScript, SQL

Frameworks: React, Flask, Django

Systems/Tools: Docker, Linux, Git, CUDA

Data/ML: NumPy, Pandas, PyTorch, Scikit-learn, TensorFlow

EXPERIENCE

Undergraduate Researcher

August 2025 - Present

University of Delaware, Thrall Lab

Newark, DE

- Built data processing pipelines on large-scale student interaction logs (NoSQL) to extract behavioral features for downstream modeling
- Fine-tuned a locally hosted LLM (Ollama) on custom datasets to extract behavioral signals, improving the classification of student engagement patterns
- Integrated model-derived features into a production-style pipeline for early risk detection across multiple checkpoints

Computer Science Ambassador

Oct 2023 – Present

University of Delaware, Department of Computer and Information Science

Newark, DE

- Coordinated 20+ technical outreach and community events per semester for prospective and current CS students

Teaching Assistant

Aug 2023 – May 2024

University of Delaware, Department of Computer and Information Science

Newark, DE

- Led lab sessions and office hours for 30+ students, teaching C programming, debugging, and systems fundamentals.

PROJECTS

Spotify Listening Analytics Dashboard — React, JavaScript, Spotify API, Flask, OAuth 2.0, Vercel

- Built a full-stack Spotify analytics dashboard using React, Flask, and REST APIs to display personalized listening insights
- Implemented client-side caching and rate-limit handling to optimize API performance and reduce redundant requests
- Developed Flask backend services for Spotify API aggregation, OAuth authentication, and analytics processing

Backtesting Engine — Python, NumPy, Pandas

- Built a modular backtesting engine in Python with layered architecture (strategy, execution, portfolio) for simulating trading systems
- Implemented order execution logic with transaction costs and slippage, modeling realistic trade behavior
- Designed reusable interfaces enabling rapid integration of new strategies and evaluation workflows
- Developed performance analytics (Sharpe ratio, drawdown) to evaluate time-series strategy performance

Non-Cartesian MRI Images Reconstruction - Python, PyTorch, OpenCV

- Reimplemented a legacy MATLAB MRI reconstruction pipeline in PyTorch using Non-Uniform Fast Fourier Transform (NUFFT) for non-Cartesian k-space data
- Parallelized MRI reconstruction pipeline using GPU acceleration (PyTorch, CUDA), reducing runtime by 40% for large-scale imaging data

LSTM Sentiment Classification Model — Python, PyTorch, LSTM

- Preprocessed 500K+ hotel reviews (Kaggle) using tokenization, padding, and vocabulary optimization for sequence modeling
- Built and trained an LSTM-based RNN for sentiment classification, incorporating dropout regularization to reduce overfitting